Roberto Albornoz Martinez
MSBA Candidate, Questrom School of Business, Boston University
BA889 Analytics Practicum - Professor Ritwick Roy
August 2026

Locked In: Mortgage Prepayment Behavior Across Rate Regimes

1. Objective

Two groups of American mortgage borrowers, separated by only two years of origination, lived through opposite interest-rate incentive environments. Loans originated in 2018, at an average rate of 4.74%, became deeply refinanceable when mortgage rates fell below 3% in 2020-21, and 78.9% of them prepaid. Loans originated in 2020, at an average of 3.19%, now sit far below a market average above 6%; only 36.9% have prepaid, and the remainder are locked in. Recent research shows that mortgage-rate differentials of this kind affect household mobility, home sales, labour reallocation, and housing-market liquidity, with the behavioural response becoming nonlinear once refinancing becomes sufficiently attractive (Batzer et al., 2024; Fonseca and Liu, 2024). This paper extends that setting to the borrower-level prepayment decision, and to a question this literature has not asked: whether a predictive model calibrated under one incentive environment transfers to another.

“Regime” in this paper denotes the incentive environment a cohort experienced - the path of its locked-in rate against market rates - not disjoint calendar periods. Both cohorts lived the same calendar history, including the 2020-21 trough, from opposite positions: the trough made the 2018 cohort deeply in-the-money, while it created the 2020 cohort at the money, leaving most of it without a meaningful refinancing window at any point. That the calendar is shared is a strength of the comparison: macroeconomic shocks are common to both cohorts, and what varies is the incentive position.

Prepayment models sit underneath the valuation of mortgage-backed securities and the management of servicing portfolios, and such models are calibrated on historical data - which means calibrated on a particular rate regime. This paper asks what happens when the regime changes: how does the rate incentive drive monthly prepayment in each regime, and does a model calibrated on one regime transfer to the other?

Using loan-level Freddie Mac data, I construct a discrete-time hazard panel for both vintages, fit a logistic baseline and a gradient-boosted tree model to each, and test every model on the regime it has never seen. The central finding: a transferred model keeps its ability to rank loans by risk but loses its ability to state how much prepayment there will be; predicted levels differ from observed levels by as much as a factor of three - and the failure is invisible on the metric that standard validation reporting leads with (AUC), while largest on the quantity that valuation runs on (calibrated levels). This project is distinct from my prior MSBA coursework, which addressed loan default classification; prepayment is a

different borrower decision, modelled here as a panel hazard with a cross-regime design rather than a cross-sectional classifier.

2. Data

The loan data is the Freddie Mac Single-Family Loan-Level Dataset (academic sample): 50,000 loans drawn from each of the 2018 and 2020 origination vintages, with monthly performance records through September 30, 2025 - 1,998,728 and 2,340,586 loan-months respectively. Each loan carries its origination characteristics (rate, balance, credit score, LTV, DTI, purpose, occupancy, term) and a monthly record of its balance, rate, delinquency status, estimated current LTV, borrower-assistance status, and termination code. Market rates come from the Freddie Mac Primary Mortgage Market Survey (PMMS): weekly 30-year fixed rates, averaged to monthly, merged onto every loan-month with no unmatched rows. Both samples are homogeneous where it matters for this design: every loan is fixed-rate, and none carries a prepayment penalty.

Voluntary prepayment is identified by termination code 01, which records early payoff in full - refinancing, home sale, or any other voluntary payoff. Throughout, "prepayment" means this voluntary full payoff; the rate incentive is one driver of it, not the only one - the decomposition is visible in the response curve of Section 4 (Figure 1), and the composition of the bundle is discussed in Section 6. Loans that ended in default-related dispositions or administrative removals (230 in 2018, 143 in 2020 - under 0.5% of each cohort) are excluded rather than treated as non-prepayments, because a loan that defaulted never made the payoff decision this paper studies. The modelled populations are therefore 49,770 and 49,857 loans, of which 78.9% and 37.0% prepaid.

Two further construction decisions shape the panel. First, the observation window is capped at 66 months of loan age - the longest window both cohorts can supply, given that the 2020 vintage is observed for at most 68 months. The cap censors 1,342 late prepayments in the 2018 cohort (3.4% of its prepayments) and 4 in 2020; censored loans remain in the panel as loans that had not prepaid within the window. Second, Freddie Mac's special missing-value codes (999 for estimated LTV and DTI, 9999 for credit score) are converted to missing throughout.

Table 1 summarises the two cohorts at origination. The comparison motivates the paper's design: the 2020 cohort is not the 2018 cohort two years later. It has a higher median credit score (768 against 754), lower original LTV (74 against 80) and DTI (34 against 37), a larger median balance ($263k against $207k) - and 70% of it consists of refinancers, against two-thirds purchase loans in 2018. The 2020 vintage is, to a substantial degree, the population that refinanced during the 2020-21 rate trough. The models adjust for these observable characteristics, but the cross-regime test necessarily evaluates a joint shift: rate environment, borrower composition, and vintage-specific selection move together - notably, much of the 2020 cohort exists in this data precisely because it refinanced into that trough. This paper does not attempt to isolate the regime's causal contribution from

composition; the joint shift is exactly what a deployed model faces when the environment turns, so the test measures deployment realism rather than causal decomposition.

Table 1. Loan characteristics at origination, by vintage

Characteristics	2018	2020
Loans	49,770	49,857
Median credit score	754	768
Median original LTV (%)	80	74
Median original DTI (%)	37	34
Median original balance ($ k)	207	263
Mean SATO (pp)	0.15	0.22
Share 30-year term (%)	84.08	72.91
Share purchase (%)	66.84	29.98
Share cash-out refinance (%)	20.87	19.29
Share owner-occupied (%)	87.33	91.68

Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

3. Methodology

Panel construction. The unit of analysis is the loan-month: every month of every modelled loan's life within the 66-month window becomes one row, carrying an indicator equal to one only in the month the loan prepaid - a discrete-time hazard structure of 1,778,314 loan-months for the 2018 cohort and 2,336,348 for 2020. This structure connects each month's decision to the conditions that month, and it handles censoring by design: loans still active at the window's end, and the late prepayers the cap excludes, contribute their months as non-events. Because a loan's final reported row describes the aftermath of payoff rather than the decision - the balance is zero and the equity estimate absent - the balance-dependent variables (unpaid balance and estimated current LTV) enter as start-of-month values, lagged one month within each loan.

Features. Each loan month carries four families of predictors, all knowable at the start of the month. The rate incentive - the loan's current rate minus the prevailing PMMS 30-year rate - is the rate component of the refinancing calculus, not its entirety. Other components enter where observed: fixed transaction costs motivate the +0.5 percentage-point threshold used below, at which the raw response in the data turns sharply upward (Figure 1), and make log balance a driver of the dollar savings; equity enters through estimated current LTV; eligibility through delinquency and assistance status. Components outside the loan file - expected tenure, idiosyncratic frictions - are unobserved, and the spread at origination (the note rate minus the market rate at the loan's first-payment month) proxies for their selection. Alongside the rate incentive itself, its path enters through burnout, the peak incentive to date, and the length of the current spell above the +0.5 threshold; origination characteristics enter through credit score, DTI, LTV, log balance, term, purpose, and occupancy; and loan age and calendar month complete the set. Missing values are filled with medians computed from training loans only, estimated-LTV missingness carries

its own indicator; and each fitted model retains its own medians for use wherever it later predicts.

Models and validation. Twenty percent of loans in each cohort are held out, split by loan so that no loan's months appear on both sides. Two models are fitted per cohort on its training loans: a logistic regression, whose coefficients, standard errors, and p-values make the economics inspectable, and a gradient-boosted tree ensemble (scikit-learn's HistGradientBoostingClassifier, default settings, fixed seed), capable of the nonlinear response the raw data displays. Standard errors for the logistic models are clustered at the loan level, since a loan's months are not independent. Event rates are left at their natural levels - 2.13% and 0.79% of training months - because the calibrated monthly probability is itself the quantity of interest, and resampling would distort it. One estimation result forced a specification difference: the delinquency and assistance indicators perfectly predict non-prepayment (no prepayment occurs in any assistance-plan month of either cohort, nor in any delinquent 2020 month), so no finite logistic coefficient exists for them; they are retained in the tree model only.

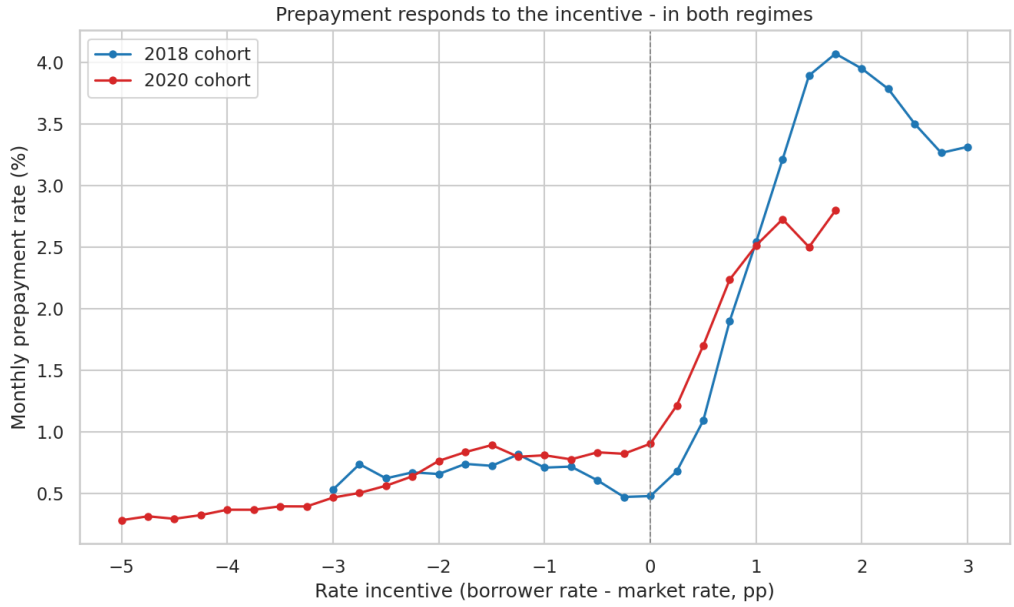
The transfer test. Each fitted model scores two test sets: its own cohort's held-out loans, which must reproduce the within-regime results exactly, and the other cohort's held-out loans - unseen loans from a different incentive environment. The two evaluations answer different questions. The within-regime holdout is cross-sectional out-of-sample: unseen loans, but the same calendar history the model trained on; it measures generalisation to new borrowers, not forecasting. The cross-regime evaluation is the distribution-shift test this paper is about. Models travel sealed: each prepares whatever data it scores using its own training medians, exactly as a frozen production model deployed into a new rate environment would. Transfer is measured on two dimensions that can fail independently: ranking (AUC - does the model still order riskier months above safer ones?) and calibration (the ratio of mean predicted to actual monthly hazard, and decile calibration curves - are the predicted levels still true?). The distinction matters because portfolio valuation depends on levels, while standard validation reporting leads with ranking.

4. Results

4.1 The raw regime contrast. Figure 1 shows the monthly prepayment rate against the rate incentive, computed directly from the loan-months of each cohort before any model. Three features organise everything that follows. Deep out of the money, both cohorts converge on roughly half a percent per month - a floor of home sales unrelated to refinancing. The response to the incentive begins near +0.5 percentage points and rises steeply: the 2018 rate nearly quadruples between +0.5 and +1.5, from 1.09% to 3.90% per month. And the two curves do not coincide - the 2020 cohort prepays more at +0.5 and substantially less at +1.5 - a first indication that the incentive response is entangled with the composition of the borrowers experiencing it. The regimes also separated the cohorts in time and in opportunity. Both cohorts' prepayment waves crest in the same calendar month - March 2021- at 5.42% and 1.99% of surviving loans, at loan ages of roughly 30 and 11 months

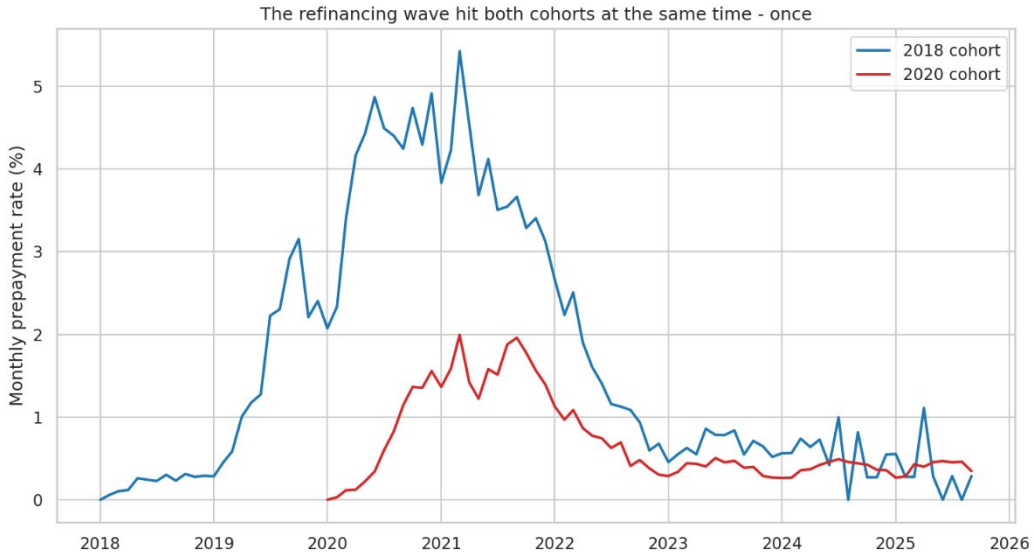
(Figure 2): timing followed market conditions, not the loan clock. The opportunity was unequal: at the trough, 99.9% of surviving 2018 loans stood above the +0.5 threshold against 44.1% of 2020 loans - most of the 2020 cohort never had an economically meaningful refinancing window at any point in the observation period. The cohorts' end states differ accordingly: 23.6% of the 2018 cohort remains active at month 66, compared with 61.6% of 2020.

Figure 1. Monthly prepayment rate by rate incentive, by cohort



Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

Figure 2. Monthly prepayment rate by calendar month, by cohort



Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

4.2 Within-regime model performance. On its own cohort's held-out loans, the logistic baseline attains an AUC of 0.740 (2018) and 0.706 (2020); the tree model reaches 0.767 and 0.755, with average precision of 0.062 and 0.031 against event-rate baselines of 0.022 and 0.008. The tree model's larger gain in the 2018 cohort is consistent with Figure 1: the floor and the elbow are precisely the nonlinearities a linear model cannot express. Discrimination in the mid-0.70s reflects the information content of a loan file - the timing of a borrower's action depends substantially on circumstances recorded in no field of the data - and a linear specification already attains 0.74. The logistic coefficients establish that the two regimes priced the same variables differently. The incentive coefficient is +0.48 per percentage point in 2018 against +0.35 in 2020. Burnout is nearly identical across regimes (-0.023 and -0.025). The spread at origination is negative in both and substantially stronger in 2020 (-0.59 against -1.39), consistent with unobserved frictions among borrowers priced above the market on day one. Credit score is highly significant in 2018 and indistinguishable from zero in 2020 - a compositional result: the 2020 cohort consists of pre-selected prime refinancers with little remaining score variation. The models fitted to the two regimes are, in substance, different models.

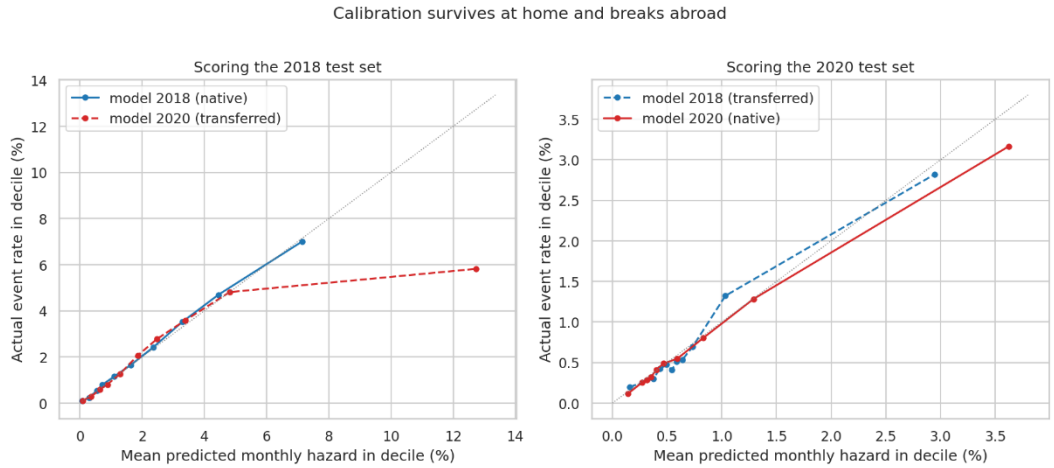
4.3 The transfer test. Table 2 reports every model evaluated on every regime. On ranking, transfer largely succeeds: every cross-regime AUC lies within 0.02-0.04 of its native benchmark (worst case 0.689 against a native 0.706). On calibration, it fails, and asymmetrically. Native models are honest, with predicted-to-actual ratios of 0.96 to 1.08. Transferred models are not: the 2018 logistic model predicts 0.62 times the true monthly hazard on 2020 data, while the 2020 logistic model predicts 3.18 times the observed monthly hazard on 2018 data. An input error of that magnitude propagates into any valuation, hedging, or servicing projection built on the model, although the size of the resulting pricing error depends on the full cash-flow path, discounting, and option structure, and is not quantified here. The tree model is more robust but not immune: 1.03 in one direction, 1.29 in the other, and Figure 3's decile calibration shows its highest-risk decile of the 2018 test set predicted at 12.7% against 5.8% observed - the riskiest segment mispriced by better than two to one. The dangerous direction is from the low-activity regime into the high-activity one.

Table 2. The cross-regime transfer test: ranking (AUC) and calibration (predicted-to-actual hazard) by model and direction.

Family	Model	Test Set	Direction	AUC	Pred/Actual
Logistic	2018	2018	Native	0.740	0.96
Logistic	2018	2020	Cross	0.689	0.62
Logistic	2020	2018	Cross	0.711	3.18
Logistic	2020	2020	Native	0.706	1.03
Gradient boosting	2018	2018	Native	0.767	0.98
Gradient boosting	2018	2020	Cross	0.719	1.03
Gradient boosting	2020	2018	Cross	0.745	1.29
Gradient boosting	2020	2020	Native	0.755	1.08

Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

Figure 3. Decile calibration of native and transferred models

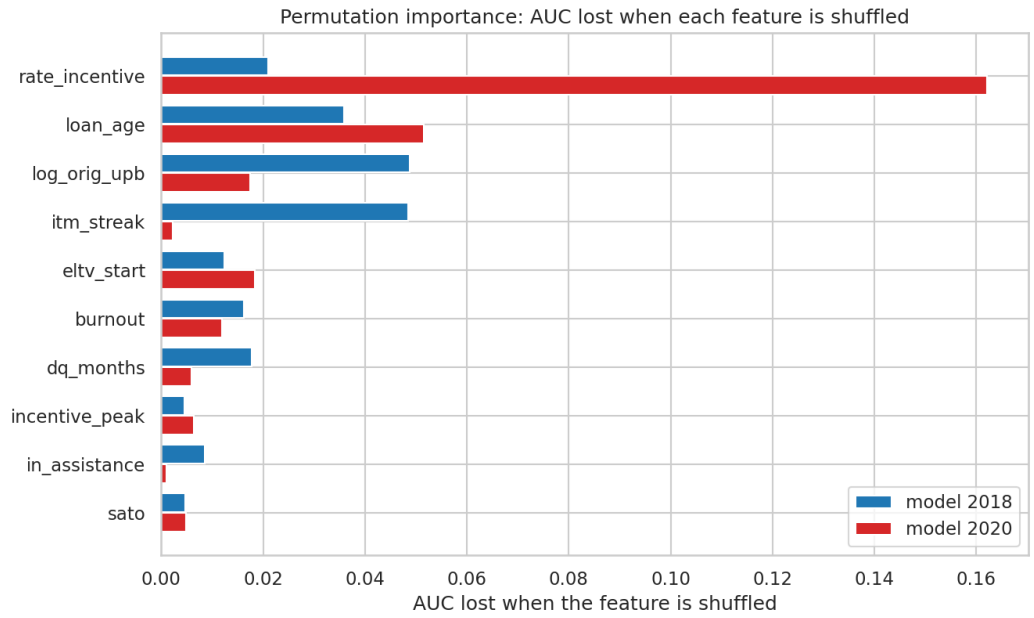


Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

4.4 The mechanism. Two diagnostics locate the failure inside the fitted models. Permutation importance (Figure 4) shows the two regimes produced different decision structures: the 2020 model depends overwhelmingly on the incentive - shuffling it costs 0.162 of AUC, more than every other feature combined - while the 2018 model distributes its reliance across loan size (0.049), the in-the-money streak (0.048), loan age (0.036), and the incentive (0.021). Where refinancing was briefly economical, identifying the window is the whole task; where it was economical for years, the task is identifying which borrowers act, and when. This is why ranking survives transfer while calibration does not.

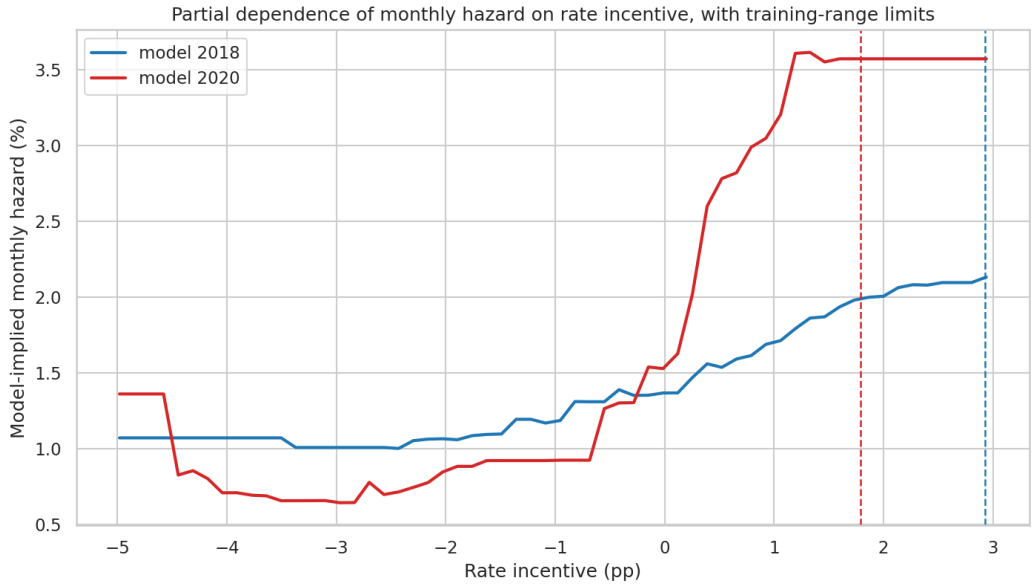
The partial-dependence curves (Figure 5) supply the calibration mechanism directly: each model's learned hazard-versus-incentive relationship ends where its training data ends. The 2020 model's curve flattens at 3.57% before its own training limit of +1.79 and answers every question beyond it with that plateau; the 2018 model's curve is still rising at +2.5. Above +1, the 2020 model's implied hazard exceeds the 2018 model's by roughly 1.5 percentage points - the direct mechanism of the 1.29x overprediction, and, in a more extreme form, of the logistic model's 3.18x. A model asked about territory outside its training regime does not fail randomly: it repeats the last relationship it learned and gives no signal that it is extrapolating.

Figure 4. Permutation importance: AUC lost when each feature is shuffled



Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

Figure 5. Partial dependence of monthly hazard on rate incentive, with training-range limits



Source: Freddie Mac Single-Family Loan-Level Dataset and PMMS; author's calculations

5. Conclusions and Recommendations

This paper set out to measure whether a prepayment model calibrated in one interest-rate regime survives deployment in the opposite one. The answer is precise: it survives in one dimension and fails in the other. A transferred model retains its ability to rank - every cross-

regime AUC lies within 0.02-0.04 of its native benchmark - but loses its ability to state how much prepayment there will be, with predicted monthly hazards differing from observed hazards by as much as a factor of three. The two dimensions are not equally visible: standard validation reporting leads with ranking metrics, while portfolio valuation and servicing economics depend on levels. A model review that checked a transferred model's AUC would conclude it still works; a portfolio priced on its predicted levels would be systematically mispriced. That gap - between the metric that gets reported and the quantity that gets priced - is the paper's central finding.

The failure has a structure, and the structure is actionable. It is asymmetric: models calibrated in the low-activity regime and deployed into the refinancing wave overpredict severely (3.18x for the logistic model, 1.29x for the tree model), because they extrapolate a steep learned response into incentive territory they never observed - the fitted curves flatten exactly where their training data ends, and give no signal that they are extrapolating. The reverse direction underpredicts (0.62x for the logistic model), because a model trained amid abundant refinancing never learned the home-sale floor that sustains prepayment when refinancing is uneconomical. The flexible model is consistently more level-robust than the linear one, but robustness is relative, not absolute: its worst decile misprices risk by better than two to one.

Three recommendations follow for practitioners who run prepayment models across changing rate environments. First, validate transferred models on regime coverage and calibration, not discrimination alone: a predicted-to-actual hazard ratio is a one-line computation, and it is the number that detects the failure AUC conceals. Second, monitor a deployed model's inputs against its training range: the damage concentrates precisely where the current incentive distribution exceeds the regime the model was fitted on, so a simple range check on the incentive is a direct early-warning signal. Third, treat a regime change as a recalibration trigger rather than trusting flexible models to bridge it: the tree model's advantage over the linear one shows that functional form helps, and its residual mispricing shows that no functional form substitutes for data from the regime being priced.

6. Limitations and Future Work

Three limitations qualify the findings, and each defines an extension. First, termination code 01 does not distinguish refinancing from home sale - both are voluntary prepayment. The floor in Figure 1 suggests the sale component is roughly half a percent per month and largely incentive-insensitive, but the two behaviours respond to different economics, and the lock-in literature concerns mobility specifically; separating them - for instance by linking payoffs to subsequent originations - is the natural refinement. Second, the 66-month observation window censors 3.4% of the 2018 cohort's prepayments, and all monthly statistics describe the loans surviving to each month: as prepayers exit, the remaining pool re-composes, a selection effect visible in the widening dispersion of the 2018 cohort's incentive distribution. The window was the price of comparing the cohorts

over identical stretches of life; a full-life robustness check on the 2018 cohort alone would test its cost. Third, the models omit local information - property location beyond state, house-price dynamics, and any borrower-level circumstances outside the loan file - which bounds discrimination and may absorb part of the compositional differences the coefficients attribute to observable characteristics.

Two extensions follow directly from the design. The 2007 vintage, held in reserve for this study, constitutes a third regime type: where 2018 borrowers had both incentive and ability, and 2020 borrowers have ability without incentive, the 2007 cohort - once rates collapsed after 2008 - had massive incentive without ability, locked out by negative equity rather than locked in by cheap debt. Testing transfer into that regime is the natural sequel, but it requires machinery this paper deliberately avoided: default exits are far too numerous there to exclude, demanding a competing-risks specification, and the estimated LTV and assistance fields are not disclosed for the relevant years. Second, the finding that no prepayment occurs in any assistance-plan month suggests modelling impairment as an explicit state - a multi-state hazard in which loans move between refinanceable and non-refinanceable status - rather than as a covariate, which would sharpen both the within-regime hazard and the transfer comparison.

7. References

Batzer, R. M., Coste, J. R., Doerner, W. M., and Seiler, M. J. (2024). "The Lock-In Effect of Rising Mortgage Rates." FHFA Working Paper 24-03. Federal Housing Finance Agency.

Fonseca, J., and Liu, L. (2024). "Mortgage Lock-In, Mobility, and Labor Reallocation." The Journal of Finance, 79(6).

Freddie Mac. Single-Family Loan-Level Dataset (academic sample) and accompanying file layout documentation. Freddie Mac, McLean, VA.

Freddie Mac. Primary Mortgage Market Survey (PMMS), weekly 30-year fixed-rate series.